# Annadhasan L3-L5 Explained From Scratch

## A slide-by-slide walkthrough for someone who missed the class

Deck: CY1010_Part4_Annadhasan_L3L4L5.pdf (64 slides). Each section explains what the slides are trying to say in plain language, then lists what is worth remembering for the MCQs. All 64 original slides are embedded in order so you can match explanation to slide.

★ = asked in 2024 Exam 2  ⚠ = slide wording is off or inconsistent

## The storyline of the whole deck

The deck answers one question: **what comes out of our cars and chimneys, and what does it do?** It walks through the pollutants one by one, and each one gets its own mini-story (source → chemistry → fate → harm → control).

| Slides | Topic | One-line idea |
|---|---|---|
| 1-17 | CO: bonding and hemoglobin | CO is a strong ligand (MO theory) that jams the $O_2$-carrying site of Hb |
| 18-21 | CO fate | OH radicals oxidise CO to $CO_2$; soil microbes also eat CO |
| 22-25 | $NO_x$ | Hot engines make NO; it becomes brown $NO_2$, then nitric acid |
| 26-38 | VOCs, engines, catalytic converter | Engines emit CO, NO and unburned fuel; the converter fixes all three |
| 39-47 | $SO_2$ and acid rain | Burning sulfur-containing coal makes $SO_2$ → sulfuric acid |
| 48-57 | Smog | Coal smog (cold, $SO_2$) vs car smog (sunny, $NO_x$ + HC → $O_3$, PAN) |
| 58-64 | Stratospheric ozone | $O_3$ forms and breaks naturally; OH, NO, Cl speed up the breaking |

## Mini-glossary (words the slides assume you know)

- **Radical:** a species with an unpaired electron (OH•, NO•, Cl•, HOO•). Very reactive; reacts until it pairs the electron up.
- **hν (photolysis):** a photon of light supplies the energy to break a bond. '$O_3$ + hν → $O_2$ + O' means light splits ozone.
- **Catalyst:** speeds a reaction and is regenerated. In ozone chemistry X is used then returned, so one X destroys thousands of $O_3$.
- **M (third body):** any inert molecule (usually $N_2$ or $O_2$) that carries away excess energy so a newly formed molecule does not fly apart.
- **Primary vs secondary pollutant:** primary = emitted directly (CO, $SO_2$, NO); secondary = made in the air ($O_3$, PAN, $H_2SO_4$, $HNO_3$).
- **Residence time:** how long a molecule stays in the atmosphere before it is removed.
- **Troposphere / stratosphere:** the lowest ~10 km (weather, smog) / the layer above (to ~50 km) that holds the ozone layer.
- **Wavenumber ($cm^{-1}$):** the unit for IR peaks; higher number = higher energy = stiffer bond.
- **ppm / ppb:** parts per million / billion by number of molecules.

# 1. Carbon monoxide: bonding and IR (slides 1-8)

## Why does a pollution lecture start with orbitals?

Because the toxicity of CO and the way CO sticks to metals both come from the **same two features of its electron structure**. The lecturer introduces these features once (slides 1-5) and re-uses them for hemoglobin (slides 6-17). If you follow the reasoning below, the slides stop being random.

### Slide 1: the MO diagram

- CO has 10 valence electrons: 4 (C) + 6 (O). Fill the molecular orbitals from the bottom: 8 electrons end up in bonding orbitals and 2 in antibonding ones. **Bond order = (8 − 2)/2 = 3**, so C≡O, and the free-molecule stretch is **2143 $cm^{-1}$** ★.
- Two orbitals do all the chemistry. **HOMO** = highest filled orbital (a lone pair on carbon). **LUMO** = lowest empty orbital, the antibonding π* (also mostly on carbon).
- **σ-donation:** CO gives its HOMO lone pair to an empty metal orbital. **π back-donation:** the metal gives electrons from a filled d orbital into CO's empty π*. Those two flows together make the M–C bond strong ('synergic bonding'). That is the drawing at the right of the slide (σ-type interaction, π-type interaction, 'electron donor / electron acceptor').
- **Consequence:** electrons going into π* (antibonding) weaken C–O. So in a metal carbonyl the C–O bond order is below 3 and ν(CO) is **lower** than 2143. The resonance picture M–C≡O ↔ M=C=O says the same: some double-bond character.
- A **node** is a plane/point where the orbital amplitude is zero. More nodes = higher energy; antibonding orbitals have an extra node between the atoms. 2024 Q8 asked the number of nodes in one particular CO orbital ★, and Q9 asked bond order (answer 3) ★.

### Slide 2: bridging modes and the ν(CO) table

- CO can sit on one metal (**terminal**) or span two or three metals (**bridging**, $\mu_2$ or $\mu_3$). The slide asks: how does the wavenumber of CO change in metal carbonyls?
- **Rule: the more electron density the metal pushes into π*, the lower ν(CO).** Table: $Ni(CO)_4$ 2057 · $Co_2(CO)_8$ 2044 · $Fe(CO)_5$ 2023 · $Mn_2(CO)_{10}$ 2013 · $Cr(CO)_6$ 2000 · $V(CO)_6$ 1976 · **$[Ti(CO)_6]^{2-}$ 1747**. The dianion has extra negative charge to give away, so it back-donates most and has the lowest ν ★ (2024 Q11).
- **Bridging CO has lower ν than terminal CO** ★ (Q10): a bridging CO accepts back-donation from two metals at once.

### Slides 3-4: IR spectra and mutual exclusion

- An IR peak appears only if the vibration **changes the dipole moment**. CO has one stretch, which is IR active (the sharp peak near 2143, shown on slide 3).
- $CO_2$ is linear and symmetric. Its **symmetric stretch** (both O move out together, ~1285 $cm^{-1}$) leaves the dipole at zero, so it is **IR inactive but Raman active**. The **asymmetric stretch** (2349) and the **bend** (666) change the dipole, so they are **IR active**.
- **Mutual exclusion principle (slide 4):** in a molecule with a centre of symmetry, a vibration cannot be both IR and Raman active. Why it is here: it explains which $CO_2$ modes you see in an IR spectrum.

### Slide 5: the worked example

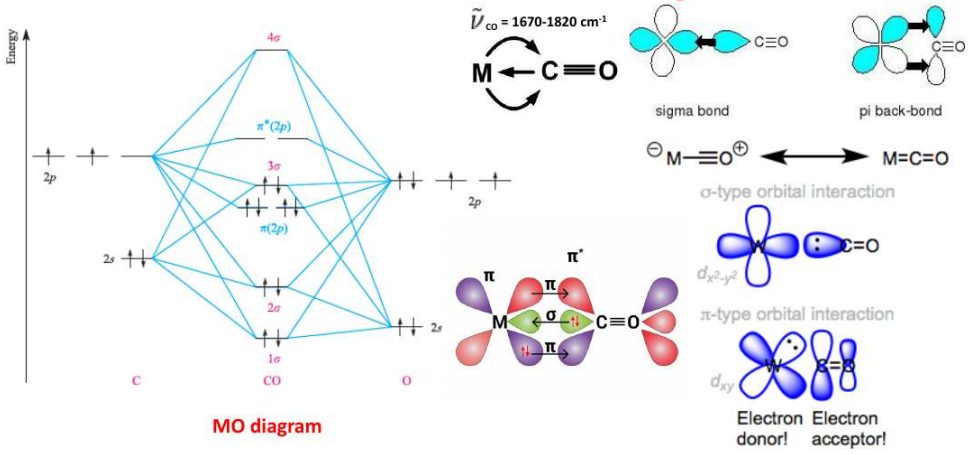
- An Fe dimer carrying both terminal and bridging COs. Terminal peaks sit at higher wavenumber (~2000), bridging at lower (~1800). One single bridging band means the two bridging COs are almost co-linear; two terminal bands means the two terminal COs are not co-linear (the slide's reading). It is a one-off illustration; the bridging-vs-terminal rule is what you need.

### Slides 7-8: IR correlation tables

- Reference tables of typical IR peak positions (C–H 2850-3300, C=O 1680-1750, O–H 3230-3550 etc.). They are background; I would not memorise them.

### Slides from the course deck (Annadhasan L3-L5)

# Molecular Orbital Theory




Bond order and bond strength **Free CO = 2143 cm$^{-1}$**

1

*Annadhasan L3-L5 · slide 1*

## Carbon Monoxide

Bridging modes of CO



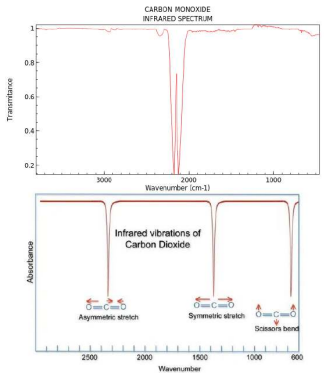
❖ How the wave number of *CO* will change in metal carbonyl complexes

| Free CO | | | | | |
|---|---|---|---|---|---|
| 2143 cm$^{-1}$ | | | | | |
| $V(CO)_6$ | $Cr(CO)_6$ | $Mn_2(CO)_{10}$ | $Fe(CO)_5$ | $Co_2(CO)_8$ | $Ni(CO)_4$ |
| 1976 cm$^{-1}$ | 2000 cm$^{-1}$ | 2013 cm$^{-1}$ | 2023 cm$^{-1}$ | 2044 cm$^{-1}$ | 2057 cm$^{-1}$ |
| $[Ti(CO)_6]^{2-}$ | | | | | |
| 1747 cm$^{-1}$ | | | | | |

⊖M—≡O⊕ ⟷ M=C=O

2

*Annadhasan L3-L5 · slide 2*




*Mutual Exclusion Principle*

For molecules with a center of symmetry, no IR active transitions are Raman active and vice versa

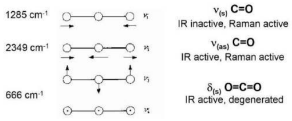
⇒*Symmetric molecules*

IR-active vibrations are not Raman-active.

Raman-active vibrations are not IR-active.

O = C = O ⟷ O = C = O

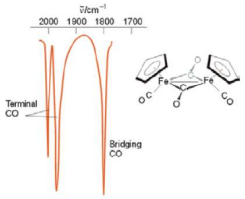
Raman active IR inactive | Raman inactive IR active




*Annadhasan L3-L5 · slide 3*

## Example

- IR spectrum shows bands in the bridging and terminal regions
- Bridging IR band is single implying that the two COs are almost co-linear
- Two terminal IR stretch bands implying that the two COs are not co-linear




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*Annadhasan L3-L5 · slide 4*

| Functional Group | Wavenumber (cm$^{-1}$) | Intensity |
|---|---|---|
| **Alkyl** | | |
| C-H | 2853-2962 | medium-strong |
| **Alkenyl** | | |
| =C-H | 3010-3095 | medium |
| C=C | 1620-1680 | variable |
| **Alkynyl** | | |
| ≡C-H | 3300 | strong |
| C≡C | 2100-2260 | variable |
| **Aromatic** | | |
| Ar-H | 3030 | variable |
| C=C | 1400-1600 | variable |
| **Alcohols, Phenols & Carboxylic Acids** | | |
| O-H (alcohols / phenols) | 3200-3500 | broad, strong |
| O-H (carboxylic acids) | 2500-3000 | broad, variable |
| **Aldehydes, Ketones, Esters, Carboxylic acids & Amides** | | |
| C=O (Aldehydes) | 1690-1740 | strong |
| C=O (Ketones) | 1680-1750 | strong |
| C=O (Esters) | 1735-1750 | strong |
| C=O (Carboxylic acids) | 1710-1780 | strong |
| C=O (Amides) | 1630-1690 | strong |
| **Amines** | | |
| N-H | 3300-3500 | medium |

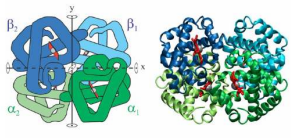
Infrared absorption data

| Bond | Wavenumber /cm$^{-1}$ |
|---|---|
| N—H (amines) | 3300 – 3500 |
| O—H (alcohols) | 3230 – 3550 |
| C—H | 2850 – 3300 |
| O—H (acids) | 2500 – 3000 |
| C≡N | 2220 – 2260 |
| C=O | 1680 – 1750 |
| C=C | 1620 – 1680 |
| C—O | 1000 – 1300 |
| C—C | 750 – 1100 |

*Annadhasan L3-L5 · slide 5*



myoglobin



hemoglobin

## Myoglobin

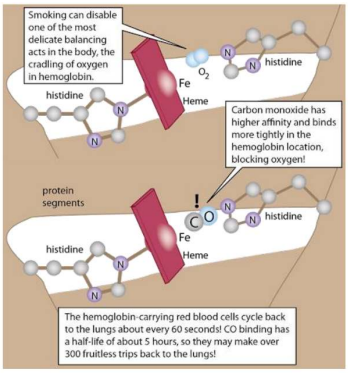
- It is a monomeric protein that has 153 amino acids
- It has 8 alpha helices
- **Myglobin** = heme group (contains iron) + Globin protein
- Have a linear chain of amino acids
- It has a globular structure

## Hemoglobin

- Quaternary structure
- Four subunits: 2 alpha and 2 beta (Adult hemoglobin)
- ***Alpha subunit:*** 141 amino acid, chromosome 16
- ***Beta, gamma and delta subunit:*** 146 amino acid, chromosome 11

- Each subunit or polypeptide has a heme prosthetic group bound
- Each Hb has four heme
- Each heme $Fe^{2+}$ binds to single oxygen molecule
- Therefore each Hb can carry four oxygen molecules

*Annadhasan L3-L5 · slide 7*




*Annadhasan L3-L5 · slide 8*

# 2. Hemoglobin, myoglobin and why CO kills (slides 6, 9-17)

## The problem in one paragraph

Your cells need $O_2$. **Hemoglobin (Hb)** in red blood cells picks it up in the lungs and drops it off in tissues. **Myoglobin (Mb)** in muscle stores it. The $O_2$ is held by **iron in a heme group**. CO binds the same iron far more tightly than $O_2$, so with CO present the iron is blocked and tissues starve.

## Slides 6-10: the structures

- **Mb**: one polypeptide (~153 aa, 8 α-helices) with **one** heme: holds one $O_2$. **Hb**: four chains (2 α + 2 β), four hemes: holds four $O_2$. The 3D ribbon pictures just show these folds; the red part is the heme.
- **Heme geometry (slide 9) ★:** Fe(II) sits in a flat porphyrin ring. Four coordination sites are taken by the four porphyrin N, the **fifth by a histidine N of the protein (proximal His)**, and the **sixth site is free for $O_2$**. When $O_2$ is attached, Fe is **six-coordinate** (2024 Q12, answer six).
- Why the protein is needed: a bare Fe(II) heme reacts with $O_2$ irreversibly (it forms dimers/oxidises). The globin **buries** the heme so two hemes cannot meet, allowing reversible binding.

## Slide 11: the key number

- $Mb(O_2)$ + CO → Mb(CO) + $O_2$ (and Hb → **carboxyhemoglobin**). The slide says the free heme group binds CO about **25,000×** better than $O_2$ ★ (2024 Q13 options: 2500, 25 thousand, 2.5 lakh, 25 lakh).
- In real Hb the preference is much smaller (~200×; another slide says 200× for HbCO). The protein pocket reduces it, as slide 17 explains below. ⚠ Read whether the question says 'free heme' or 'hemoglobin'.

## Slides 12-14: derivatives and function

- Slide 13: Mb stores $O_2$, aids diffusion; Hb transports it from lungs to tissue.
- Slide 14 lists the 'normal' derivatives (oxy-, deoxy-, carbamino-Hb) and 'abnormal' ones (**met-Hb** with $Fe^{3+}$ that cannot carry $O_2$; **carboxy-Hb** with CO; **sulf-Hb**, green; acid hematin). Glycated Hb (HbA1c) is the sugar-linked form used to monitor diabetes. The picture of green blood is sulfhemoglobinemia.

## Slides 15-16: how $O_2$ and CO sit on iron, and the curves

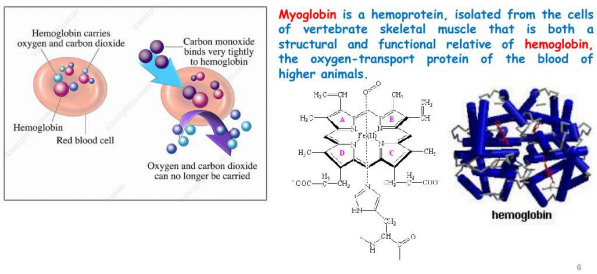
- Space-filling pictures: $O_2$ sits bent on Fe between the proximal His (below) and distal His (above). CO is drawn in the same pocket.
- **Saturation curves (slide 16):** Mb is a **hyperbola** (grabs $O_2$ strongly even at low pressure; good storage). Hb is **S-shaped (sigmoidal)** because its four sites cooperate: binding one $O_2$ makes the next easier; it loads in the lungs (high $pO_2$) and unloads in tissue (low $pO_2$). At lower pH (6.8 vs 7.6, e.g. working muscle) Hb gives up $O_2$ more readily (the curve shifts right; the Bohr effect).

## Slide 17: why is CO weaker in the protein than on bare heme?

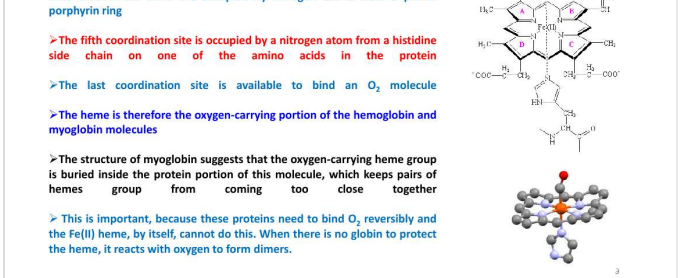
- The protein pocket makes room for $O_2$, which binds bent, but **CO prefers to bind linearly** (Fe–C–O in a line). The distal His blocks that: **steric clash**, so CO must bind at an angle, which is a weaker bond. Also $Fe^{2+}$ is low spin. The net effect is that the protein trims CO's advantage from ~25,000× to ~200×, but 200× is still enough to be poisonous.
- The CO symptom slides (in L1-L2): 35 ppm headache; 200 ppm loss of judgement; 6400 ppm death in minutes.

Exam hooks: heme is six-coordinate in oxy-Hb ★; CO affinity 25,000× (free heme) ★; Hb sigmoidal, Mb hyperbolic; $Fe^{3+}$ = met-Hb; steric clash reason; HbCO half-life ~5 h.

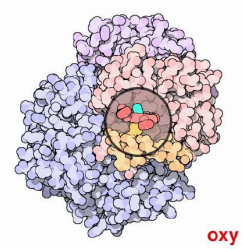
## Slides from the course deck (Annadhasan L3-L5)




*Annadhasan L3-L5 · slide 6*




*Annadhasan L3-L5 · slide 9*




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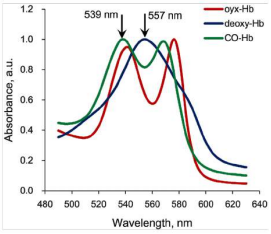
*Annadhasan L3-L5 · slide 10*

**Preferential Binding of CO and deactivation of Hb and Mb**

$Mb(O_2) + CO \rightleftharpoons Hb(CO) + O_2$

Carboxyhemoglobin

Binding affinity of free Heme group for CO is about 25,000 times than $O_2$

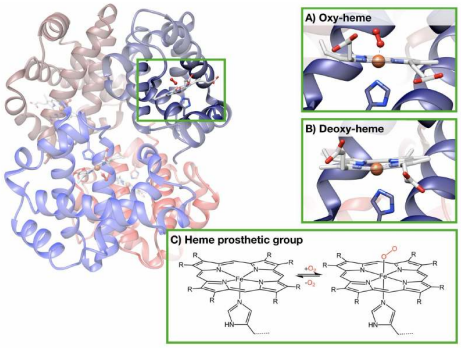



**Oxygen Binding to Mb and Hb**

- **Oxymyoglobin** - oxygen bearing myoglobin
- **Deoxymyoglobin** - oxygen-free myoglobin
- In oxymyoglobin, six ligands are coordinated to the ferrous ion in octahedral symmetry
- Oxygen is coordinated between the iron and the imidazole sidechain of His-64

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*Annadhasan L3-L5 · slide 11*




*Annadhasan L3-L5 · slide 12*

**MYOGLOBIN**

- Myoglobin (Mr 16,700; abbreviated Mb) is a relatively simple oxygen-binding protein found in almost all mammals, primarily in muscle tissue
- Mb is a single polypeptide chain with one molecule of heme. Human Mb contains 152 amino acids with a molecular weight of 17,500 Daltons. One molecule of Mb can combine with 1 molecule of oxygen.
- In the muscles, the oxygen is taken up by Mb for the sake of tissue respiration.

**Myoglobin and Hemoglobin**

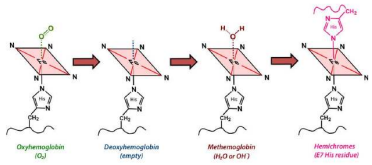
- **Myoglobin (Mb)**
  - Increases $O_2$ solubility in tissues (muscle)
  - Facilitates $O_2$ diffusion
  - Stores $O_2$ in tissues
- **Hemoglobin (Hb)**
  - Transports $O_2$ from lungs to peripheral tissues (erythrocytes)

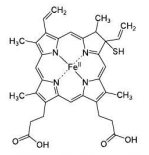
**Derivatives of hemoglobin**

- **Oxyhemoglobin** (oxyHb) = Hb with $O_2$
- **Deoxyhemoglobin** (deoxyHb) = Hb without $O_2$
- **Methemoglobin** (metHb) contains $Fe^{3+}$ instead of $Fe^{2+}$ in heme groups
- **Carboxyhemoglobin** (HbCO) – CO binds to $Fe^{2+}$ in heme in case of CO poisoning or smoking. CO has 200x higher affinity to $Fe^{2+}$ than $O_2$.
- **Carbaminohemoglobin** ($HbCO_2$) – $CO_2$ is non-covalently bound to globin chain of Hb. $HbCO_2$ transports $CO_2$ in blood (about 23%).
- **Glycohemoglobin** (HbA1c) is formed spontaneously by nonenzymatic reaction with Glc. People with DM have more HbA1c than normal (> 7%). Measurement of blood HbA1c is useful to get info about long-term control of glycemia.

*Annadhasan L3-L5 · slide 13*

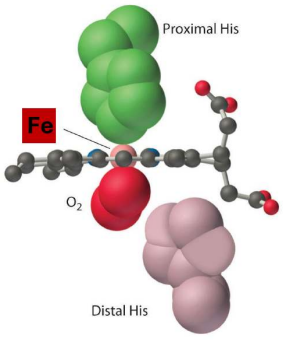
| Normal Derivatives | Abnormal Derivatives |
|---|---|
| 1. Oxyhemoglobin | 1. Methemoglobin |
| 2. Deoxyhemoglobin | 2. Carboxyhemoglobin |
| 3. Carbaminohemoglobin | 3. Sulfhemoglobin |
| | 4. Acid hematin |

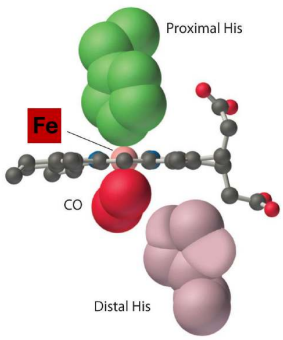




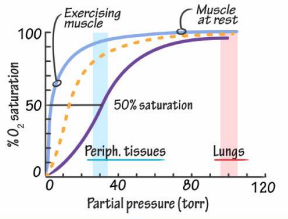

*Annadhasan L3-L5 · slide 14*

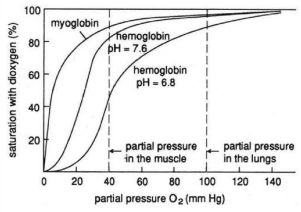



*Annadhasan L3-L5 · slide 15*

**Dissociation Curves**



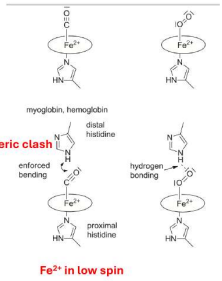




Oxygen saturation curves of myoglobin and hemoglobin at different pH values

| Myoglobin | Hemoglobin |
|---|---|
| Higher $O_2$ affinity | Cooperative binding = sigmoidal curve |
| Responds to muscle's $O_2$ needs | Responds to $O_2$ availability |

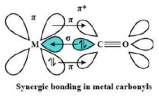
16

*Annadhasan L3-L5 · slide 16*




Protein part allows easier access to small molecules like dioxygen to bind with Fe atoms

Linear coordination of CO is not possible




**Bonding is weak when CO is bound in angular fashion**

17

*Annadhasan L3-L5 · slide 17*

# 3. What happens to CO in the air (slides 18-21)

- **Slide 18:** CO is removed mostly by **oxidation**. The three oxidants listed: **OH radical** (the 'detergent of the atmosphere' ★), nitrate radical $NO_3$, and ozone. ⚠ The slide says this happens in the stratosphere; in reality OH oxidation of CO is mainly tropospheric. Answer with the slide wording unless an option explicitly says troposphere.
- **Slide 19:** CO + HO• → $CO_2$ + H•. The reaction is fast and nearly independent of temperature, so CO's **residence time is short: 2 weeks to 3 months**. (HOO• is called dioxidanyl; this is naming trivia.)
- **Slide 20:** where do OH and HOO• come from? $O_2$ + H → HOO•; HOO• + NO → OH + $NO_2$; 2HOO• → $H_2O_2$ + $O_2$; $H_2O_2$ + hν → 2OH. And from ozone: $O_3$ + hν (<310 nm) → O + $O_2$, then O + $H_2O$ → 2OH. So sunlight + ozone + water makes the OH that eats CO.
- **Soil sink:** soil microbes metabolise CO; soil accounts for **~40%** of total CO consumption. Slide 21 lists what controls soil uptake (water content, temperature, organic matter, pH, soil type). Humic acid formula $C_{187}H_{186}O_{89}N_9S_1$ is only an average composition, not examinable in practice.

## Slides from the course deck (Annadhasan L3-L5)

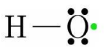
### Carbon Monoxide

**Fate of Atmospheric CO**

Most of the CO molecules are removed from the atmosphere by oxidation (natural) in the stratosphere.

**The most important oxidizing species in the air**

❑ Hydroxyl radical OH

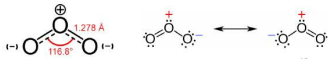


❑ Nitrate radical $NO_3$



'detergent of the atmosphere'.

❑ Ozone molecule $O_3$




18

*Annadhasan L3-L5 · slide 18*

### Carbon Monoxide

**Fate of Atmospheric CO**

It is generally believed that CO is removed from the atmosphere by a reaction with hydroxyl radical.

**Oxidized by reaction with hydroxyl radical**

• CO + HO• → $CO_2$ + H (stratosphere)




Systematic IUPAC name
**Dioxidanyl**

- CO is oxidized to $CO_2$ in the stratosphere, where it migrates *via* convection, turbulence, and mixing
- The reaction is fast and independent of temperature; thanks to it, the residence time of CO in the atmosphere is relatively short, from 2 weeks to 3 months.

19

*Annadhasan L3-L5 · slide 19*

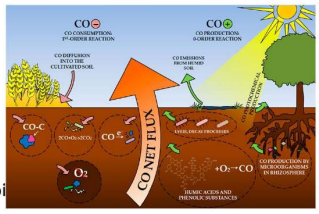
### Carbon Monoxide

**Production of hydroperoxyl radical**

• $O_2$ + H → HOO•

**Additional reactions**

• HOO• + NO → HO• + $NO_2$
• HOO• + HOO• → $H_2O_2$ + $O_2$
• $H_2O_2$ + *hν* → HO•
$O_3$ + *hν* (sun) → O + $O_2$ (wavelength <310 nm)
O + $O_2$ + *M* → $O_3$ + *M*
O + $H_2O$ → 2OH 3%




CO Fate in Soi

- Soil is considered one of the main sinks of atmospheric CO, responsible for 40% of total consumption
- Soil is a sink for CO
- Soil microorganisms metabolize CO

Humic acid has the average chemical formula $C_{187}H_{186}O_{89}N_9S_1$

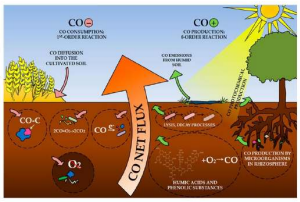
20

*Annadhasan L3-L5 · slide 20*

**The most important soil parameters**

- Water Content
- Temperature
- Organic matter content
- pH
- Soil type

The depth of CO consumption horizon, and CO concentration in the gas phase




Even small changes in this balancing between CO production and soil uptake can severely impact tropospheric chemistry

*Annadhasan L3-L5 · slide 21*

# 4. Nitrogen oxides (slides 22-25)

- **Slide 22:** $NO_x$ means NO + $NO_2$. $N_2$ and $O_2$ do not react at normal temperature, but in an engine or power-plant flame the temperature is so high that $N_2 + O_2 \rightarrow 2NO$, and then $2NO + O_2 \rightarrow 2NO_2$. The pie chart: transportation 49.1% and stationary fuel combustion 43.2% (together ~92%).
- **Slide 23:** NO is a radical (an unpaired electron, NO•) and is quickly turned into $NO_2$ in air. Bacterial breakdown of N-containing organic matter in soil is a natural source. $NO_2$ is removed as nitric acid through OH radicals: **$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$**; much of this happens inside water droplets in clouds, giving acid rain.
- **Slide 24:** $HNO_3$ also reacts with ammonia to form ammonium nitrate particles ($HNO_3 + NH_3 \rightarrow NH_4NO_3$), which fall out as acid deposition.
- **Slide 25:** $NO_2$ is a **red-brown, toxic, sharp-smelling gas ★** (2024 Q17: the coloured one is $NO_2$); it irritates eyes, inflames lung tissue, causes emphysema, and is important because it drives photochemical smog. The slide defines erythema, emphysema, melanogenesis, atopic sensitisation (medical terms, low priority).

## Slides from the course deck (Annadhasan L3-L5)

### Nitrogen Oxides ($NO_x$)

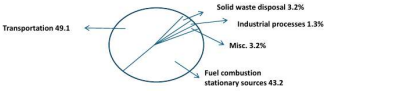
❖ Nitrogen dioxide ($NO_2$) is the major pollutant in the atmosphere and is formed from nitric oxide (NO). Collectively nitrogen oxides are designated as $NO_x$

➢ Source of $NO_x$

➢ Anthropogenic $NO_x$ molecules enter the atmosphere from the combustion of fossil fuels by automobiles, aircraft, and power plants.

➢ At normal atmospheric temp. $N_2$ and $O_2$ do not react. However, at very high temp. atmospheric $N_2$ reacts with oxygen to give a series of complex reactions. Two gases combine to give NO

$N_2 + O_2 \longrightarrow 2NO \qquad 2NO + O_2 \longrightarrow 2NO_2$




22

*Annadhasan L3-L5 · slide 22*

### Nitrogen Oxides ($NO_x$)

➢ When NO is released, it combines rapidly with atmospheric oxygen to form $NO_2$. NO has an unpaired electron and can be written as NO•

• The bacterial decomposition of nitrogen-containing organic matter in soil is another natural source of $NO_x$

❖ Fate of Atmospheric $NO_x$

1. Regardless of its source, $NO_2$ is ultimately removed from the atmosphere as nitric acid and nitrates in dust and rainfall. In a series of complex reactions involving hydroxyl radicals, $NO_2$ combines with water vapor to form nitric acid. The simplified overall reaction is

$$4\,NO_2 + 2\,H_2O + O_2 \longrightarrow 4\,HNO_3$$

2. Much of the nitric acid in the atmosphere is formed within aqueous aerosols. If the weather conditions are right, the aerosols coalesce into larger droplets in clouds and the result is acid rain.

23

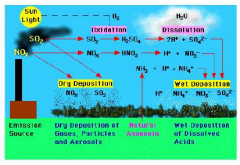
*Annadhasan L3-L5 · slide 23*

### ❖ Fate of Atmospheric $NO_x$

❖ Some of the nitric acid formed reacts with ammonia and metallic particles in the atmosphere to form nitrates

$$HNO_3 + NH_3 \longrightarrow NH_4NO_3$$

✓ Nitrates dissolve in rain and snow or settle as particles. The combined fallout contributes to acid deposition




24

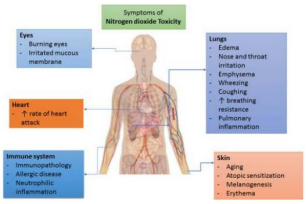
*Annadhasan L3-L5 · slide 24*

### Effect of $NO_x$ on Human Health and Environment

➢ $NO_2$ is red brown toxic gas and has unpleasant acrid odor. It can cause irritation of the eyes, inflammation of lung tissue, and emphysema.

➢ $NO_x$ is a serious health problem because of its role in the formation of secondary pollutants associated with photochemical smog.

❖ Most of the research has concentrated on reducing automobile emission by means of the catalytic converter




**Emphysema** is a lung disease that damages the alveoli in your lungs. The primary symptom of emphysema is shortness of breath.

**Erythema** is redness of the skin caused by injury or another inflammation-causing condition

**Melanogenesis** is the production of the melanin pigments

**Atopic sensitization** is defined as a positive allergen-specific serum immunoglobulin E (IgE) test or skin prick test to any food or inhalant allergens. It is an important risk factor for the development of asthma and allergic diseases

*Annadhasan L3-L5 · slide 25*

# 5. VOCs, gasoline and engines (slides 26-34)

- **VOCs** are volatile carbon compounds that evaporate easily. Slide 27: the biggest human source is the petroleum industry/transport (60%); gasoline vapours escape when you pump fuel, when tanks are filled, and as unburned fuel in exhaust. Nature makes VOCs too: **terpenes** give pine, eucalyptus and sandalwood their smell (slide 28; general formula $(C_5H_8)_n$; structures are limonene, myrcene, linalool etc.).
- **Slide 29: gasoline** is a mixture of hydrocarbons (alkanes 4-8%, isoalkanes 25-40%, aromatics 20-50%, benzene 0.5-2.5%). **Tetraethyl lead** was added to stop *knocking* (uneven burning).

## The four-cycle (ordinary car) engine, slides 30-31

- Four steps per cycle: intake, compression, power, exhaust. Inside the cylinder: **limited $O_2$ → some carbon ends as CO**; **very high T and P → $N_2$ + $O_2$ → NO**; **some fuel never burns → unburned HC** in the exhaust. That is why cars emit all three pollutants.
- **Air/fuel ratio:** the ideal (stoichiometric) mass ratio is **14.7 : 1** ★. The slide's trade-off: *fuel-rich* (lower ratio) = easier start, lower $NO_x$ and HC, but more CO and worse mileage; *air-rich* (higher ratio) = better efficiency, harder start, more $NO_x$. The ideal ratio is a compromise.

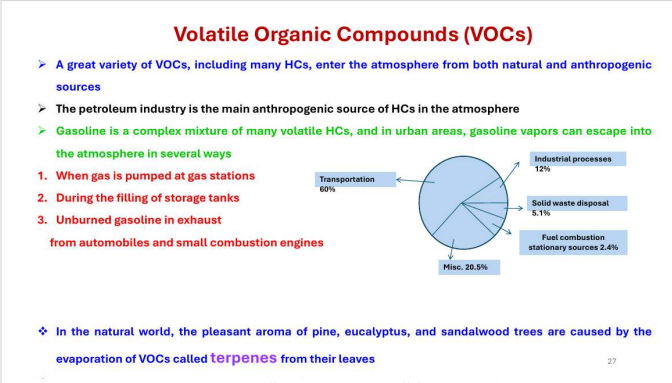
## The two-cycle engine, slides 32-33

- Used in scooters, mopeds, snowmobiles. Cheaper and simpler: no valves, one power stroke per two strokes. But the fresh fuel-air mix enters while exhaust leaves through the same stroke, so **25-30% of the fuel leaves unburned**. It runs on a fuel-oil mix with motor oil : gasoline = **1 : 40**; the oil ($C_{18}$-$C_{25}$, heavy) is not vaporised and burned well. Combustion is cooler than in a four-cycle engine, so it makes less NO.
- Slide 33 compares marine outboards: two-cycle CO 165 g, $NO_x$ 0.3 g, HC 89 g versus four-cycle 127, 0.7, 7. Two-cycle emits **more than 12×** the hydrocarbons, but only about a third as much $NO_x$ (cooler flame).

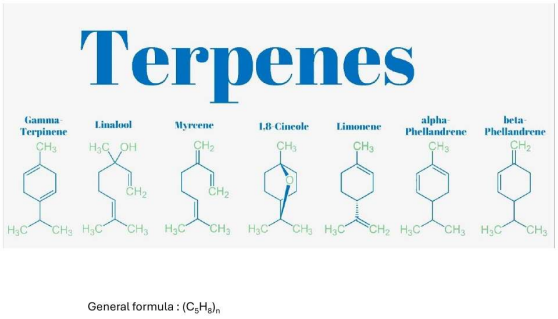
## Slide 34: emission standards

- Since 1975 (US) all new cars must carry a catalytic converter. The table shows allowed g/mile falling from HC 10.6, CO 84, $NO_x$ 4.1 (before 1970) to 0.41, 3.4, 0.4 (2000): today's car emits about **95% less**, even though miles driven nearly doubled.

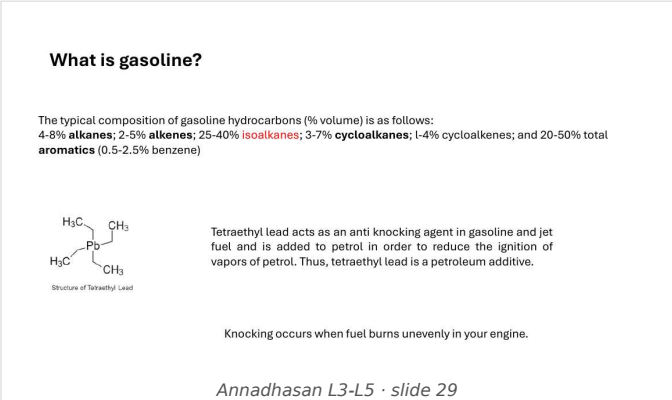
## Slides from the course deck (Annadhasan L3-L5)



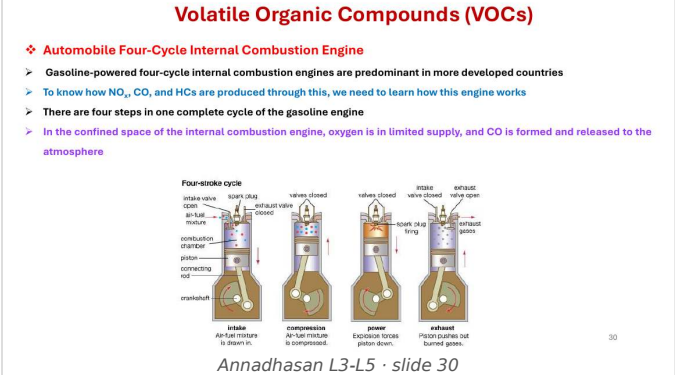

*Annadhasan L3-L5 · slide 27*




*Annadhasan L3-L5 · slide 28*




*Annadhasan L3-L5 · slide 29*




*Annadhasan L3-L5 · slide 30*

## Volatile Organic Compounds (VOCs)

- Automobile Four Cycle Internal Combustion Engine
  - In the internal combustion engine, temp and pressure are very high, and NO is formed from $N_2$ and $O_2$. The unburned HCs are emitted with the burned gas as pollutant VOCs
  - The air/fuel ratio has a dramatic effect on the emission of pollutants from the four-cycle engine
  - During the engine tune-up, adjustment of the carburetor or the fuel injection system can greatly reduce the emission of pollutants.
  - The ideal stoichiometric air/fuel ratio is 14.7:1
  - Adjustment to a more fuel-rich (lower air/fuel) ratio makes the car easier to start and decreases the emission of NOx and HCs; unfortunately, it also decreases the fuel efficiency, and the car will get fewer Km per liter. The emission of CO is also increased as air is less available
  - Adjustment to a more air-rich (higher air/fuel) ratio will make the car harder to start; although it increases fuel efficiency, the emission of $NO_x$ is increased
  - The ideal ratio is a compromise between fuel efficiency and the emission of the lowest collective amount of pollutant

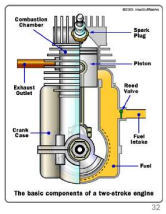
31

*Annadhasan L3-L5 · slide 31*

## Volatile Organic Compounds (VOCs)

- Gasoline Powered Two-Cycle Engine
- A two-cycle engine is used to power motor scooters, mopeds, and snowmobiles. It is less complicated than the four-cycle engine and also less expensive.
- It takes fuel, releases exhaust in the same stroke, and emits from 25% to 30% of the fuel consumed as unburned HCs.
- The two-cycle engine has no valves, and that power is produced in one of two steps rather than one in four steps; this means the two-cycle engine runs faster and hotter than the four-cycle engine

1. In two-cycle engine motor oil: gasoline 1: 40 ratio
2. Motor oil has $C_{18}$ to $C_{25}$ HCs i.e., higher molecular weight
3. In the two-cycle engine, combustion takes place at a lower temperature than the four-cycle engine
4. Due to higher molecular weight, motor oil is not vaporized and burned efficiently like gasoline
5. Two step cycle requires that the exhaust gases from combustion leave the cylinder through the exhaust port while the fresh air/fuel mixture for the next power stroke is simultaneously enters the cylinder from the side arm



The basic components of a two-stroke engine

32

*Annadhasan L3-L5 · slide 32*

## Volatile Organic Compounds (VOCs)

- Comparison

1. A comparison of the emission from marine outboard two- and four-cycle engines of the same horsepower was made. The result of those tests show that the exhaust of a two-cycle engine contained more than 12 times the amount of HCs than a four-cycle engine of the same power

| Type of Marine outboard Engine | CO(g) | $NO_x$(g) | HC(g) |
|---|---|---|---|
| Two cycle Engine | 165 | 0.3 | 89 |
| Four Cycle Engine | 127 | 0.7 | 7 |

*Source: Juttner F. D. et al., Water Research* **1995**, *29*, 1976-1982



33

*Annadhasan L3-L5 · slide 33*

## Automobile Pollutants

- Motor vehicles are a major source of CO, $NO_x$, and volatile HCs
- Since 1975, when all new cars in the USA were required by law to be equipped with a catalytic converter, emissions of those pollutants have been reduced significantly
- The table shows that today's car emits 95% less pollutants than pre-1970 vehicles despite the fact that the number of miles traveled has almost doubled in the last 20 years

| Emission Standards for Light duty Vehicles | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Federal Standards | | | | California Standards | | | |
| Year | HCs | CO | NOx | Evaporative HCs | HCs | CO | NOx | Evaporative HCs |
| <1970 | 10.6 | 84 | 4.1 | > 45 | 10.6 | 84 | 4.1 | >45 |
| 1970 | 4.1 | 34 | - | - | 4.1 | 34 | - | 2 |
| 1975 | 1.5 | 15 | 3.1 | 2 | 0.9 | 9 | 2.0 | 2 |
| 1980 | 0.41 | 7.0 | 2.0 | 2 | 0.39 | 9 | 1.0 | 2 |
| 1985 | 0.41 | 3.4 | 1.0 | 2 | 0.39 | 7 | 0.4 | 2 |
| 1990 | 0.41 | 3.4 | 1.0 | 2 | 0.39 | 7 | 0.4 | 2 |
| 1993 | 0.41 | 3.4 | 1.0 | 2 | 0.25 | 3.4 | 0.4 | 2 |
| 2000 | 0.41 | 3.4 | 0.4 | 2 | 0.25 | 3.4 | 0.4 | 2 |

*All values reported in gms per mile except for evaporative HCs which are expressed as gms per tests*

34

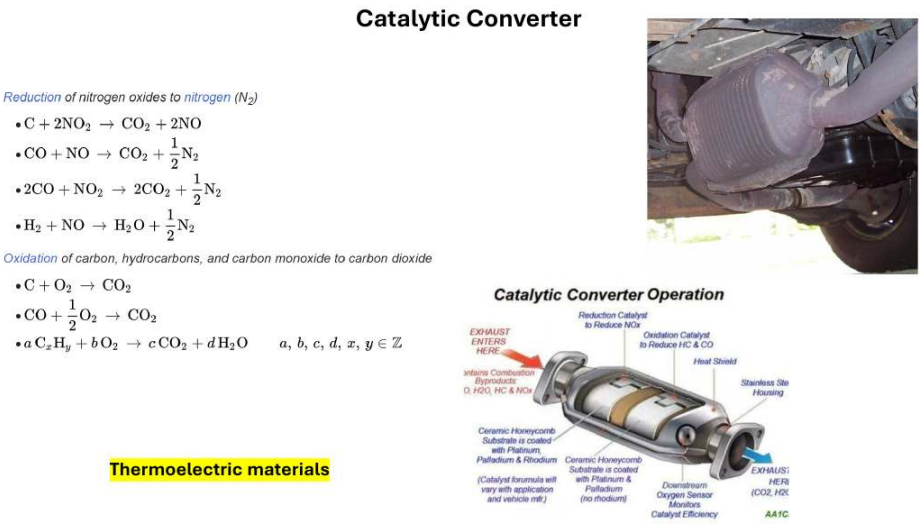
*Annadhasan L3-L5 · slide 34*

# 6. The catalytic converter (slides 26, 35-38)

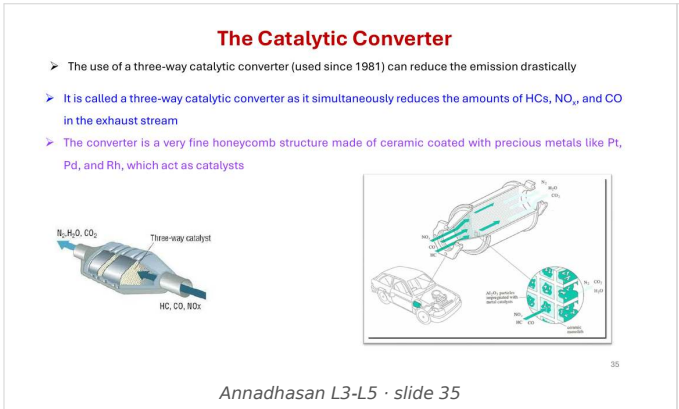
- **Slide 35:** the **three-way** converter (in use since 1981) removes HC, $NO_x$ and CO simultaneously. It is a ceramic **honeycomb** (huge surface area) coated with **Pt, Pd and Rh**. The slide also names the market leaders (BASF, CDTi, Continental, Eberspächer, Faurecia) in slide 36; ignore for exams.
- **Slide 37: two chambers in series. Chamber 1 (reduction), Rh ★:** unburned HC reacts with water to make $H_2$ ($HC + H_2O \rightarrow H_2 + CO$), and that $H_2$ reduces NO: **$2NO + 2H_2 \rightarrow N_2 + 2H_2O$**. **Chamber 2 (oxidation), Pt/Pd:** air is injected to supply $O_2$, converting CO and HC to harmless $CO_2$ and $H_2O$: $2CO + O_2 \rightarrow 2CO_2$; $HC + 2O_2 \rightarrow CO_2 + 2H_2O$.
- **Overall (the line to remember): $2NO + 2CO \rightarrow N_2 + 2CO_2$**: NO is reduced and CO is oxidised at once.
- Why air is added *after* the Rh chamber (my reasoning, not on the slide): if $O_2$ were present in chamber 1 it would burn the $H_2$ and CO that are needed to reduce NO.
- **Slide 38:** an example HC oxidation: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$. The engine must run at 14.7:1 for the converter to work, and the converter's efficiency is nearly zero at room temperature (it needs to warm up). Pt, Rh, Pd are all FCC metals (a surface-chemistry note).
- **Slide 26** (placed early in the deck) lists the reduction reactions ($C + 2NO_2 \rightarrow CO_2 + 2NO$; $CO + NO \rightarrow CO_2 + ½N_2$; $2CO + NO_2 \rightarrow 2CO_2 + ½N_2$; $H_2 + NO \rightarrow H_2O + ½N_2$) and oxidation reactions ($C + O_2 \rightarrow CO_2$; $CO + ½O_2 \rightarrow CO_2$; general HC burning) with a labelled cutaway diagram. The label 'thermoelectric materials' is a stray note.

Exam hooks: Rh = $NO_x$ reduction to $N_2$ ★ (2024 Q5 and Q19; Q19 called it 'two-way'); Pt/Pd = oxidation; A/F 14.7:1; ~0 efficiency when cold.

## Slides from the course deck (Annadhasan L3-L5)




*Annadhasan L3-L5 · slide 26*




*Annadhasan L3-L5 · slide 35*




*Annadhasan L3-L5 · slide 36*

## The Catalytic Converter

- The catalytic converter has two chambers in succession. As the gases enter, **Rh** catalyzes the reduction of $NO_x$ to nitrogen gas by hydrogen generated at the surface of the **Rh** catalyst by the reaction of $H_2O$ and unburned HCs

$$HCs + H_2O \longrightarrow H_2 + CO\,; \qquad 2NO + 2H_2 \longrightarrow N_2 + 2H_2O$$

- Then air is injected into the exhaust stream to introduce oxygen and **CO** is oxidized to $CO_2$ in the presence of **Pt** and **Pd** catalyst

$$2\,CO + O_2 \longrightarrow 2\,CO_2\,; \quad HC + 2O_2 \longrightarrow CO_2 + 2\,H_2O$$

- Thus, overall reaction for reduction of NO and oxidation of CO can be written as

$$2\,NO + 2\,CO \xrightarrow{\text{Rh, Pt, Pd Cat}} N_2 + 2\,CO_2$$

37

*Annadhasan L3-L5 · slide 37*

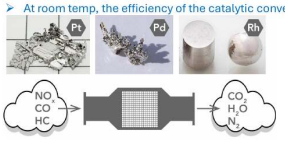
## The Catalytic Converter

- Oxidation of a typical gasoline HC, Octane occurs as

$$2\,C_8H_{18} + 25\,O_2 \xrightarrow{\text{Pt, Pd Cat}} 16\,CO_2 + 18\,H_2O$$

- Automobiles using catalytic converters must have their air/fuel ratio set as 14.7:1 to get optimum results.

- At room temp, the efficiency of the catalytic converter is nearly zero









Surface chemistry

Pt, Rh, and Pd (FCC)

38

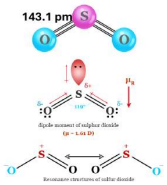
*Annadhasan L3-L5 · slide 38*

# 7. Sulfur dioxide and acid rain (slides 39-47)

- **Slide 39:** $SO_2$ (S–O 143.1 pm) is the main cause of acid rain. Fossil fuels give 70% of it and industry ~23%. Coal and oil contain sulfur (the slide writes FeS), which burns to $SO_2$. Also, $H_2S$ from anaerobic decay of sulfur-containing organic matter oxidises: $2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O$.
- **Slide 40:** volcanoes are a natural source: **Mt. Pinatubo, June 1991, 25 million tons $SO_2$** → sulfuric acid aerosols. **Residence time 10-30 days**, set by conversion to sulfate and by dry deposition.
- **Slides 41-42: the chemistry.** $SO_2 + ½O_2 \rightarrow SO_3$; $SO_3 + H_2O \rightarrow H_2SO_4$. Or $SO_2$ dissolves in droplets: $SO_2 + H_2O \rightleftharpoons H_2SO_3$ (sulfurous acid), then $H_2O_2$ or $O_3$ oxidises $HSO_3^-$ to $HSO_4^-$ (sulfate). Acid becomes concentrated near cloud bases where pH falls to ~3 (orange-juice acidity).
- **Slide 43, oxoacids of sulfur ★ (image slide).** Names and formulas: **sulfurous $H_2SO_3$**; **sulfuric $H_2SO_4$**; **thiosulfuric $H_2S_2O_3$** (one O replaced by S); **dithionous $H_2S_2O_4$**; **dithionic $H_2S_2O_6$**. Memory aid (mine): 'di-' means two S joined to each other; the '-ous' acid has fewer O ($O_4$) than the '-ic' acid ($O_6$). The 2024 paper asked the name of $H_2S_2O_6$ (answer: dithionic).
- **Slide 44:** acid rain eats marble and limestone: the Taj Mahal photo shows yellowing. (Chemistry: acid + $CaCO_3$ → soluble or flaky calcium salts; not on the slide.) **Slide 45:** $SO_2$ is colourless with a sharp odour; irritates eyes and airways (children vulnerable) and damages barley, alfalfa, cotton and wheat.
- **Slide 46: control.** Remove sulfur before burning (cleaning coal) or capture $SO_2$ in the chimney (cheaper, common). The standard method is **flue gas desulfurisation (FGD)**: pass flue gas through a slurry of limestone ($CaCO_3$) or dolomite. **$2SO_2 + 2CaCO_3 + O_2 \rightarrow 2CaSO_4 + 2CO_2$**. A basic carbonate neutralises an acidic gas.
- **Slide 47:** carbon disulfide $CS_2$ (colourless liquid, ether-like odour, bp 46.3 °C, S=C=S 155.26 pm) burns to $CO_2$ + $SO_2$; shown with a furnace setup photo. Low priority.

## Slides from the course deck (Annadhasan L3-L5)

### Sulfur Dioxide




- The release of $SO_2$ into the atmosphere is the primary cause of acid rain
- Fossil fuel combustion accounts for 70% of the emission
- Industrial sources contribute approximately 23% of $SO_2$
- Coal oil and all other fossil fuels naturally contain some sulfur (FeS).

- When sulfur containing coal is burned, the sulfur is oxidized to $SO_2$: $S + O_2 \longrightarrow 2SO_2 + 2H_2O$
- $H_2S$ (produced as an end product of anaerobic decomposition of sulfur containing organic matter by microorganism) enter to the atmosphere to form $SO_2$

$$2H_2S + 3O_2 \longrightarrow 2SO_2 + 2H_2O$$

39

*Annadhasan L3-L5 · slide 39*

### Sulfur Dioxide





- Volcanic eruptions are another more localized natural source of $SO_2$; e.g., the eruption of Mt. Pinatubo in the Philippines in June 1991 contributed 25 million tons of $SO_2$ into the atmosphere, where it was converted into sulfuric acid aerosols

- **Fate of atmospheric $SO_2$: Acid Rain**
- The residence time of $SO_2$, controlled not only by chemical conversion to sulfate but also by dry deposition, is 10-30 days.

- $SO_2$ in the atmosphere reacts with oxygen to form $SO_3$, which then readily reacts with water vapor or water

*Annadhasan L3-L5 · slide 40*

### The fate of atmospheric $SO_2$: Acid Rain

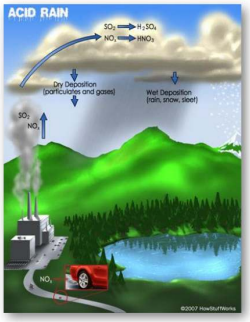
$$SO_2 + OH^{\cdot} \longrightarrow HSO_3^{\cdot}$$
$$HSO_3^{\cdot} + O_2 \longrightarrow SO_3 + HOO^{\cdot}$$
$$SO_3 + H_2O \longrightarrow H_2SO_4\ (g)$$
$$H_2SO_4 + H_2O \longrightarrow H_2SO_4\ (aq)$$

- Sulfuric acid in the atmosphere becomes concentrated near the base of clouds where the pH level is as low as 3 (pH of orange juice)
- Some of the atmospheric $SO_2$ dissolves if there is significant water in the air to form sulfurous acid

$$SO_2(g) + H_2O\ (aq) \longrightarrow H_2SO_3(aq)$$

- The dissolved $SO_2$ is oxidized by trace amount of $H_2O_2$ and Ozone that are also present in the aerosol droplets to sulfate ions ($SO_4^{2-}$)
- $HSO_3^- + H_2O_2 \longrightarrow H_2O + HSO_4^-$

$$HSO_3^- + O_3 \longrightarrow O_2 + HSO_4^-$$

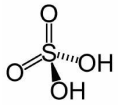



41

*Annadhasan L3-L5 · slide 41*

### The fate of atmospheric $SO_2$: Acid Rain

$$SO_2 + OH^{\cdot} \longrightarrow HSO_3^{\cdot}$$
$$HSO_3^{\cdot} + O_2 \longrightarrow SO_3 + HOO^{\cdot}$$
$$SO_3 + H_2O \longrightarrow H_2SO_4\ (g)$$
$$H_2SO_4 + H_2O \longrightarrow H_2SO_4\ (aq)$$



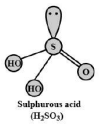
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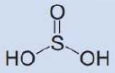
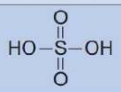
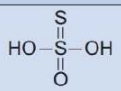
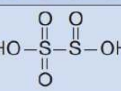
$H_2SO_4$ *vs* $H_2SO_3$




42

*Annadhasan L3-L5 · slide 42*

## Oxoacids of Sulfur

| Name | Molecular Formula | Structure |
|---|---|---|
| Sulphurous acid | $H_2SO_3$ |  |
| Sulphuric acid | $H_2SO_4$ |  |
| Thiosulphuric acid | $H_2S_2O_3$ |  |
| Dithionous acid | $H_2S_2O_4$ |  |
| Dithionic acid | $H_2S_2O_6$ |  |

*Annadhasan L3-L5 · slide 43*

## Effect of Acid Rain




Original color of Taj Mahal

Taj Mahal after the constant effect of acid rain

44

*Annadhasan L3-L5 · slide 44*

## Effect of $SO_2$ on Human Health and Environment

- $SO_2$ is a colorless, toxic gas with a sharp, acidic odour
- Exposure to it causes irritation of eyes, respiratory passages, and aggravates symptoms of respiratory disease. Children are susceptible to its effects
- $SO_2$ is also harmful to plants. Crops such as barley, alfalfa, cotton, and wheat are particularly adversely affected


Barley


Alfalfa


Cotton


Wheat

45

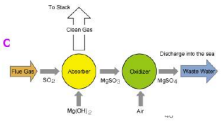
*Annadhasan L3-L5 · slide 45*

## Effect of $SO_2$ on Human Health and Environment

### ❖ How to control $SO_2$ emissions

1. Sulfur can be removed from coal before combustion
2. $SO_2$ can be removed from the smokestack after combustion but before it reaches to atmosphere

- The second approach is cheaper and most common
- The most commonly used method is flue gas desulfurization (FGD), in which sulfur-containing compounds are washed out by passing the Chimney (flue) gases through a slurry of water mixed with finely ground limestone ($CaCO_3$) or dolomite [$Ca•Mg(CO_3)_2$] or both.
- On heating the basic $CaCO_3$ with acidic $SO_2$ in the presence of $O_2$ to form C




$$2SO_2 + 2CaCO_3 + O_2 \longrightarrow 2CaSO_4 + 2CO_2$$

**Flue Gas:** a mixture of gases produced by the burning of fuel or other materials in power stations and industrial plants and extracted via ducts.

*Annadhasan L3-L5 · slide 46*

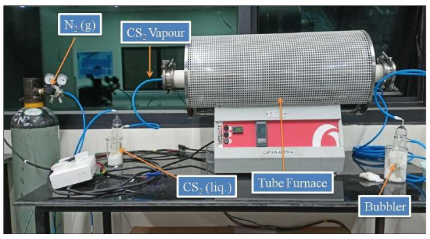
S=C=S

155.26 pm

a colorless liquid with an ether-like odor

Boiling point: 46.3 °C

$CS_2(s) + O_2(g) = CO_2(g) + SO_2(g)$




*Annadhasan L3-L5 · slide 47*

# 8. Smog (slides 48-57)

## Two very different smogs (slides 48-50)

| | Industrial smog | Photochemical smog |
|---|---|---|
| When/where | Winter, cold and damp, coal-burning cities (London) | Hot, sunny summer, traffic-heavy cities (Los Angeles) |
| Ingredients | Fly ash, soot, $SO_2$ (+ some VOCs) | $NO_x$ + hydrocarbons + sunlight |
| Look | Grey/black fog, visibility of a few yards | Yellow-brown haze: $NO_2$ colour |
| Harm | $SO_2$ + particles: deadly combination | $O_3$, PAN, aldehydes: eyes, lungs, plants |

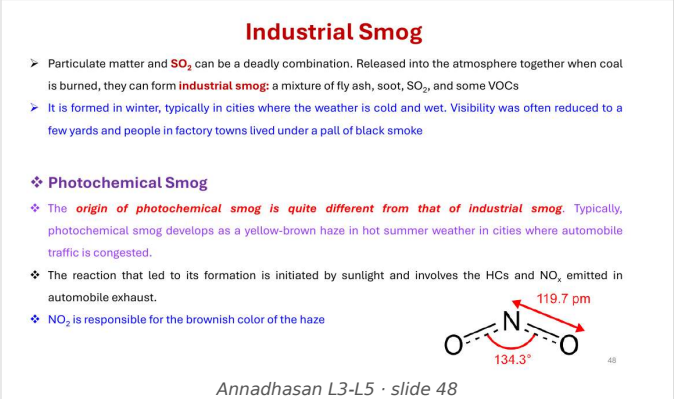
Slide 49: $NO_2$ turns into colourless $N_2O_4$ below −11.2 °C and back to $NO_2$ when warmer (this is also why $NO_2$, not $N_2O_4$, is the brown gas at room temperature). Slide 50 is photos of the two smogs.

## How photochemical smog forms: the radical chain (slides 51-56)

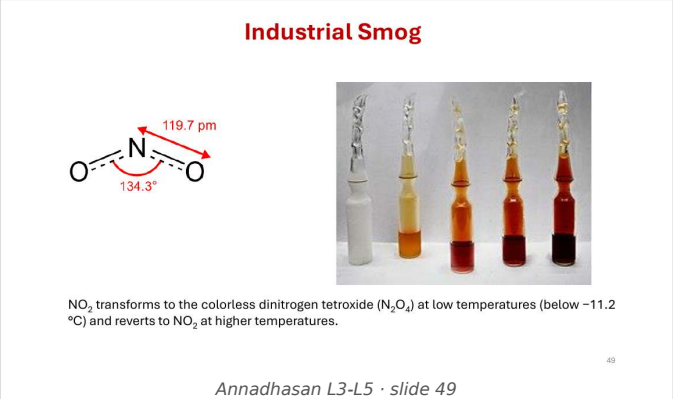
Read this as one story with five steps. Each step feeds the next, and the loop makes more radicals than it uses, which is why a small amount of $NO_x$ can generate a lot of smog.

- **Step 1: sunlight splits $NO_2$ (slide 51).** UV is all absorbed in the stratosphere, but $NO_2$ absorbs the light that does reach the ground: $NO_2 + h\nu \rightarrow NO + O$. Then $O + O_2 \rightarrow O_3$, and $O_3 + NO \rightarrow NO_2 + O_2$.
- **Step 2: make OH.** The $O_3$ is split by light: $O_3 + h\nu \rightarrow O_2 + O^*$. The oxygen atom grabs hydrogen from water: **$O + H_2O \rightarrow 2OH\bullet$**. So **one $NO_2$ molecule gives two OH radicals** (OH has seven electrons: one unpaired). Termination: radicals react with each other into water-soluble products that rain out, so the concentration does not run away.
- **Step 3: OH attacks unburned fuel (slides 52-53).** $RCH_3 + OH\bullet \rightarrow R\bullet + H_2O$; $R\bullet$ picks up $O_2$ to give $RO_2\bullet$; $RO_2\bullet + NO \rightarrow RO\bullet + NO_2$ (this $NO_2$ makes more $O_3$); eventually aldehydes and $HO_2\bullet$ form, and $HO_2\bullet + NO \rightarrow OH\bullet + NO_2$. Every step makes a radical, so the slide counts **four OH radicals per hydrocarbon reacted** ★ (2024 Q18, options 10,000 / 100 / 10 / ... fit 'a small number', answer 4 if present). It is **catalytic**: a few radicals produce a large amount of product.
- **Step 4: which hydrocarbons react fastest? (slides 54-55).** OH pulls off an H atom; the weaker the C–H bond, the faster. Bond energies (kJ/mol): methane 427; ethane 406; propane 393; methanol 393; benzene 427; **toluene's $CH_2$–H 326**. Tertiary C–H breaks more easily than secondary than primary because the resulting radical is more stable. Benzene's C–H is strong because the carbon is $sp^2$ (more s-character). Toluene/xylene react fast because the radical is stabilised by the adjacent ring. **Alkenes** react even faster because OH *adds* to the C=C instead of pulling off H; the products (acetaldehyde, formaldehyde, 2-hydroxypropanol) go on to make more pollutants.
- **Step 5: the nasty end products (slide 56).** Aldehydes continue to react with OH, $O_2$ and $NO_2$ and end up as **PAN** (peroxyacetyl nitrate, $CH_3COO_2NO_2$), the component that causes major **eye irritation** ★ (2024 Q22). PANs are stable in cool air. The pollutant made in the biggest quantity, and the most harmful in the troposphere, is **ozone** (irritates eyes and nose, toxic to plants). Do not confuse it with the helpful ozone of the stratosphere.
- **Slide 57: indoor pollution.** Sealed homes and offices with poor ventilation can be worse than outdoors; the lead-pipe photo is an example of an indoor source.

### Slides from the course deck (Annadhasan L3-L5)

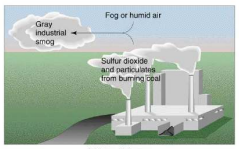



*Annadhasan L3-L5 · slide 48*

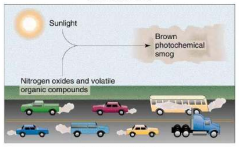



*Annadhasan L3-L5 · slide 49*

# Industrial Smog



(a) Industrial smog



(b) Photochemical smog





50

*Annadhasan L3-L5 · slide 50*

# Production of Hydroxyl Radicals

- By the time sunlight reaches the surface of the Earth, all of the light energy UV light has been absorbed in the stratosphere
- $NO_2$ is the only automobile emission that is capable of absorbing visible light that reaches the Earth's surface
- $NO_2$ + sunlight (< 320 nm) $\longrightarrow$ NO + O ; O + $O_2$ $\longrightarrow$ $O_3$
- $O_3$ + NO $\longrightarrow$ $NO_2$ + $O_2$

- The ozone produced absorbs light in the blue region of the visible spectrum (< 320 nm) and photo dissociates: $O_3 \longrightarrow O_2 + O$
- The oxygen atom produced (having six electrons) reacts with water vapor in the atmosphere and abstracts a hydrogen atom to produce hydroxyl radical (seven electrons)
- $O + H_2O \longrightarrow 2OH^\bullet$ By this way one $NO_2$ molecule produces two hydroxyl radicals
- The concentration of hydroxyl radicals does not continue to increase out of control because there are termination reactions that remove it from the troposphere. It can react with other radical species in the troposphere

51

*Annadhasan L3-L5 · slide 51*

# Production of Hydroxyl Radicals

$^\bullet OH + {}^\bullet NO_2 \longrightarrow HNO_3$

$^\bullet OH + HOO^\bullet \longrightarrow H_2O + O_2$

$2\,{}^\bullet OOH \longrightarrow H_2O_2 + O_2$

❖ The products are very soluble in water and are removed from troposphere during precipitation

- Unburned HCs in automobile exhausts ($RCH_3$) react with hydroxyl radicals to form a number of secondary pollutants, including HC radicals $RO_2^\bullet$. This radical then reacts with NO to form aldehydes and the hydroperoxide radical $HO_2^\bullet$

$RCH_3 + {}^\bullet OH \longrightarrow {}^\bullet RCH_2 + H_2O$

$RCH_2^\bullet + O_2 \longrightarrow RCH_2OO^\bullet$ (Peroxy alkyl radical)

$RCH_2OO^\bullet + NO \longrightarrow RCH_2O^\bullet$ (Alkoxy radical)

$RCH_2O^\bullet + O_2 \longrightarrow RCHO + {}^\bullet OOH$ (Hydroperoxyl radical)

$HOO^\bullet + NO \longrightarrow NO_2 + {}^\bullet OH$

52

*Annadhasan L3-L5 · slide 52*

# Production of Hydroxyl Radicals

❖ Each step in this reaction produces a radical. The overall reaction can be summarized as

$RCH_3 + 2\,O_2 + 2\,NO \longrightarrow RCHO + 2\,NO_2 + H_2O$ (1)

$2\,NO_2 + 2\,H_2O \longrightarrow 2\,NO + 4\,{}^\bullet OH$ (2)

$RCH_3 + 2\,O_2 + H_2O \longrightarrow RCHO + 4{}^\bullet OH$

❖ This reaction produces four hydroxyl radicals for every HC reacted. This is a catalytic reaction. A very small number of radicals can produce a large amount of product through the production of four radicals per cycle

## Reaction of Hydroxyl Radicals with HCs

❖ Abstraction of Hydrogen

- Hydroxyl radicals will react with certain unburned HCs from the automobile exhaust depending on the number and type of C-H bonds in the HCs

53

*Annadhasan L3-L5 · slide 53*

## Reaction of Hydroxyl Radicals with HCs

| Compound | Bond | Energy KJ/mol |
|---|---|---|
| Methane | $CH_3$-H | 427 |
| Ethane | $CH_3$-$CH_2$-H | 406 |
| Propane | $CH_3$-$CH_2$-$CH_2$-H | 393 |
| Methanol | $HOCH_2$-H | 393 |
| Benzene | $H_5C_6$-H | 427 |
| Toluene | $H_5C_6H_2C$-H | 326 |

- The dissociation reaction is R-H $\longrightarrow$ R• + H•
- The bond dissociation energy depends on how the R radical is stabilized $3^o > 2^o > 1^o$ i.e., a tertiary C-H bond more easily dissociate than a secondary C-H bond
- For benzene, the C-H has high dissociation energy as the aromatic carbon is $sp^2$, which mean $sp^2$ C-H bonds are much stronger than aliphatic C-H bond due to higher *s*-character
- Higher reaction rates (low dissociation energy) are observed for toluene or xylene as the $C_6H_5CH_2$• can be stabilized by adjacent aromatic ring

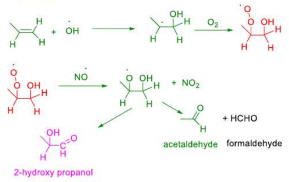
54

*Annadhasan L3-L5 · slide 54*

## Reaction of Hydroxyl Radicals with HCs

❖ **Addition to the double bond**

- **The reaction of hydroxyl radicals with alkenes proceeds at even a faster rate than the hydrogen abstraction for HCs**
- **This reaction is not a hydrogen atom extraction but rather an addition of the hydroxyl radical to the double bond**




- **The $NO_2$ produced in this reaction can go on to make more ozone, and the products of this addition reaction i.e., acetaldehyde, formaldehyde, and 2-hydroxypropanol all go on to form other pollutants in secondary smog-forming reactions**

55

*Annadhasan L3-L5 · slide 55*

# Secondary Smog Forming Reactions

- The products formed in the previous reactions can undergo further reactions to the troposphere. The following reaction sequence takes place for all aldehydes formed. Acetaldehyde is used as example

$CH_3CHO + {}^\bullet OH \longrightarrow CH_3CO^\bullet + H_2O$

$CH_3CO^\bullet + O_2 \longrightarrow CH_3COOO^\bullet$

$CH_3COOO^\bullet + {}^\bullet NO_2 \longrightarrow CH_3COOONO_2$

PAN

$H_3C-\overset{O}{\overset{\|}{C}}-O-O-NO_2$

Peroxyacetyl nitrate

- PAN is the component of smog that causes major eye irritation. PANs are relatively stable molecules and have long lifetimes in cooler air
- Ozone, aldehyde, and PANs all contribute to the harmful effects of photochemical smog, but ozone the pollutant produced in the greatest quantity causes the most serious problems in the troposphere
- Ozone in the stratosphere protects us from damaging UV radiation from the Sun
- Ozone is a powerful oxidizing agent and causes irritation in the eyes and nasal passages
- Ozone is also very toxic to plants

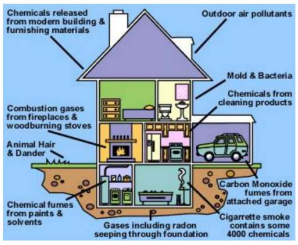
56

*Annadhasan L3-L5 · slide 56*

# Indoor Pollution

- One can expect to be safer from air pollutants indoors, but in today's well-sealed homes and offices, this is often not the case
- In buildings where there is little or no circulation or fresh air, pollutants may accumulate to a dangerous level

**Lead pipes**




57

*Annadhasan L3-L5 · slide 57*

# 9. Stratospheric ozone (slides 58-64)

## Slide 58: ozone as sunscreen and the Dobson unit ★

- The ozone layer filters harmful UV. Amount of ozone is reported in **Dobson units**: **1 DU = 2.7×10$^{16}$ ozone molecules per cm$^2$** of column (2024 Q23 options differed only in area unit and exponent: cm$^2$). (A good intuition: if all the ozone above you were compressed at surface conditions, 1 DU is a layer 0.01 mm thick.)
- Average ozone was >270 DU in the past and **293.4 DU in 2000**. Over Antarctica it falls to about **220 DU**, which is called the **ozone hole**. The unit is named after G. M. B. Dobson who built the first ground instrument.

## Slides 59-61: the natural ozone cycle (Chapman cycle)

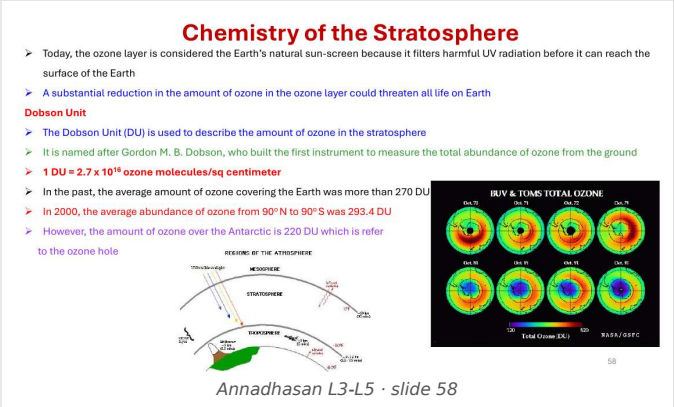
- **Ozone is made:** $O_2$ absorbs UV with **λ < 240 nm** and splits: $O_2 + h\nu \rightarrow 2O$ (slow, ΔH = 498 kJ/mol). Then $O + O_2 + M \rightarrow O_3 + M +$ heat (fast). M is a third molecule ($N_2$) that carries the heat away. That is why the layer warms.
- **Ozone is destroyed naturally:** $O_3$ absorbs UV of 220-330 nm and splits into $O_2 + O$; also $O + O_3 \rightarrow 2O_2$. Production and destruction balance: a dynamic equilibrium. Maximum concentration is at **20-30 km**.
- **Slide 60 table of UV bands:** far UV (10-240 nm) absorbed by $O_2$/$N_2$ in the thermosphere; **UV-C 250-290 nm absorbed by $O_3$** in the stratosphere; **UV-B 290-320 nm** by $O_3$ (stratosphere and troposphere); **UV-A 320-380 nm by $NO_2$** in the polluted troposphere; visible 400-750 nm reaches the Earth. Ozone absorbs 220-330 nm, with the strongest absorption around 250-260 nm ★.

## Slides 60-64: the catalytic destroyers

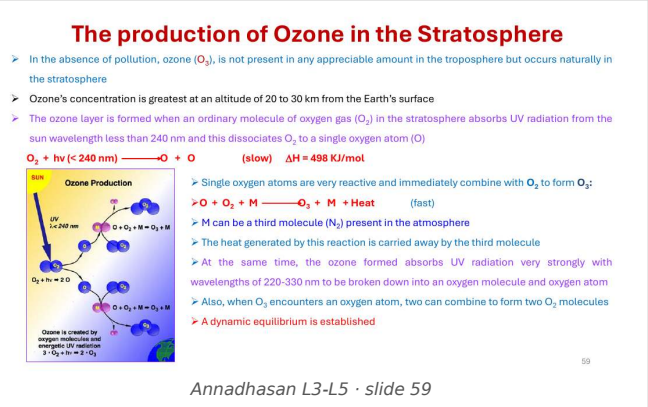
- Ozone is 'meta-stable': it decomposes slowly. Some species X speed that up by pulling an O atom off: **(i) $X + O_3 \rightarrow XO + O_2$; (ii) $XO + O \rightarrow X + O_2$; net: $O_3 + O \rightarrow 2O_2$** (slide 61). X is regenerated in step (ii), so it is a **catalyst** (it lowers the activation energy and atomic O is needed to regenerate it). X can be **OH, NO, Cl or Br**.
- **OH cycle (slide 62):** OH comes from $O + H_2O$ or $O + CH_4$, or from water photolysis ($H_2O + h\nu \rightarrow H + OH$). It causes nearly **half** of ozone loss in the lower stratosphere; one OH can run as many as **40 cycles**.
- **NO cycle (slide 63):** car NO is converted to $HNO_3$ and rained out before it reaches the stratosphere. Instead, **$N_2O$** (much less reactive) gets up there. Above 30 km: $N_2O + h\nu \rightarrow N_2 + O^*$; below 30 km: $N_2O + O^* \rightarrow 2NO\bullet$. Then NO acts as X. **One $NO_x$ can destroy ~10,000 ozone molecules.**
- **Chlorine cycle (slide 64):** CFCs (refrigerants, aerosol propellants, circuit-cleaning solvents) and bromine **halons** (fire extinguishers) were prized because they are non-toxic and non-flammable. They are so unreactive that they last >100 years in the troposphere, drift up, and are split by UV into Cl•. **Rowland and Molina (1974)** predicted this. **One Cl• can destroy ~100,000 $O_3$** before being carried down, and **bromine is more efficient than chlorine**.

Chain-length ranking to remember: **Cl (100,000) > NO (10,000) > OH (40)**. Also: 220-330 nm $O_3$ absorption, 240 nm $O_2$ cutoff, 498 kJ/mol, 20-30 km, 2.7×10$^{16}$ molecules/cm$^2$, 293.4 and 220 DU.

### Slides from the course deck (Annadhasan L3-L5)




*Annadhasan L3-L5 · slide 58*




*Annadhasan L3-L5 · slide 59*

## The production of Ozone in the Stratosphere

| Common names for the UV spectrum | | | |
|---|---|---|---|
| l(nm) | Name | Species absorbing | Location |
| 10-240 | Far UV | $O_2$, $N_2$ | Thermosphere |
| 250-290 | UV-C | $O_3$ | Stratosphere |
| 290-320 | UV-B | $O_3$ | Stratosphere, troposphere |
| 320-380 | UV-A | $NO_2$ | Polluted troposphere |
| 400-750 | Visible | Many | Earth's Surface |

### Catalytic destruction of Ozone

- Ozone is a very reactive molecule and is termed a meta-stable molecule i.e. it decomposes slowly in contact with a molecule of another gas
- There are a number of species that react efficiently by abstracting an oxygen atom from the ozone molecule. Say "X" designates the reactive species. Thus, the reaction steps are

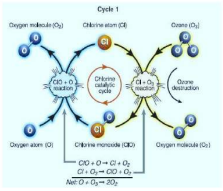
  i) $X + O_3 \longrightarrow XO + O_2$;  ii) $XO + O \longrightarrow X + O_2$
- These are the additional steps for ozone destruction compared to the steps discussed earlier

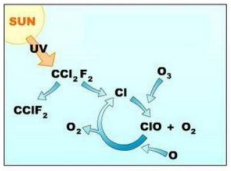
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## The production of Ozone the Stratosphere

- So, the sum of above two reactions is $O_3 + O \longrightarrow 2O_2$
- The overall reaction does not contain the X species, because X is not consumed in the reaction. It acts as a catalyst for the destruction of ozone
- "X" lowers the activation energy and atomic oxygen is required to regenerate X
- The catalytic species "X" can be i) Hydroxyl radical, ii) nitric oxide, iii) Chlorine , and iv) bromine atoms




DESTRUCTION OF OZONE BY CFC

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## Catalytic destruction of Ozone

### Hydroxyl Radical Cycle

- Hydroxyl radical (•OH) can be produced by two different photochemical processes. The first is hydrogen abstraction with either water or methane

  i) $O + H_2O \longrightarrow 2\,{}^{\bullet}OH$;  ii) $O + CH_4 \longrightarrow {}^{\bullet}OH + {}^{\bullet}CH_3$
- The second is photolysis of **water** $H_2O + hv \longrightarrow H^{\bullet} + {}^{\bullet}OH$
- Hydroxyl radical is responsible for nearly one-half of the total ozone destruction in the lower stratosphere

$$^{\bullet}OH + O_3 \longrightarrow {}^{\bullet}OOH + O_2$$
$$^{\bullet}OOH + O \longrightarrow {}^{\bullet}OH + O_2$$

One •OH radical can have as many as 40 cycles i.e., 40 ozone molecule can decompose

Net reaction $O + O_3 \longrightarrow 2O_2$

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## Catalytic destruction of Ozone

### Nitric Oxide Cycle

- Auto mobiles and truck engines release large quantities of nitric oxide (NO) into the troposphere
- Nitric oxide is converted to $NO_2$ and eventually to $HNO_3$ which fall down as rain fall before reaching to stratosphere
- $N_2O$ is much less reactive than NO and reaches to stratosphere
- Above 30 km, the $N_2O$ can absorb high energy photons to produce molecular nitrogen and an excited oxygen atom (O*)  $N_2O + hv \longrightarrow N_2 + O^*$
- Below 30 km in the stratosphere, the excited state oxygen reacts with the $N_2O$ to produce NO

  ii) $N_2O + O^* \longrightarrow 2\,NO^{\bullet}$
- NO can act as X in the catalytic process

1 NOx radical can catalyze the destruction of 10,000 ozone molecules

$NO + O_3 \longrightarrow NO_2 + O_2$
$NO_2 + O \longrightarrow NO + O_2$
Net $O + O_3 \longrightarrow 2O_2$

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## Catalytic destruction of Ozone

### The Chlorine Cycle

- The major sources of stratospheric chlorine and bromine have been the anthropogenic **chlorofluoro carbons (CFCs)** and bromine-containing halons (haloalkanes)
- CFCs used as refrigerants, propellants for aerosol spray, and solvents for cleaning electric circuit
- Halons are used in commercial fire extinguisher system
- CFCs and halons are nontoxic and nonflammable, which make them superior to other gases that were used for the same application
- CFCs are so unreactive they don't break down when released and can persist in the troposphere for more **than 100 years.** Over time, air currents carry them into the stratosphere
- In 1974, two chemists, F. S. Rowland and Mario Molina, predicted that when exposed to UV radiation in stratosphere, CFCs would break down to form chlorine radical (Cl•)
- Each chlorine radical involved in a chain reaction has the potential to destroy 100,000 molecules of ozone before winds carry it back to the troposphere
- Bromine-containing compounds are more efficient in destroying the ozone

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# 10. One-page recap

| Topic | Remember |
|---|---|
| CO bonding | Bond order 3; σ-donation + π back-donation; ν(CO) free 2143, lower in complexes; bridging < terminal; $[Ti(CO)_6]^{2-}$ lowest (1747) |
| Hb/Mb | Fe(II) heme, 5th site His, 6th site $O_2$ (six-coordinate); Hb sigmoidal, Mb hyperbolic; CO ~25,000× on free heme (~200× in Hb); steric clash/angular CO |
| CO fate | $OH + CO \rightarrow CO_2 + H$; residence 2 weeks-3 months; soil ~40%; OH = detergent of the atmosphere |
| $NO_x$ | $N_2 + O_2 \rightarrow 2NO$ at high T; $NO_2$ red-brown; $4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$ |
| Engines | 4-stroke: CO (little $O_2$), NO (hot), HC (unburned); A/F 14.7:1; 2-stroke 25-30% unburned, oil:gasoline 1:40, >12× HC |
| Converter | Rh reduces $NO_x \rightarrow N_2$; Pt/Pd oxidise CO, HC; air injected; $2NO + 2CO \rightarrow N_2 + 2CO_2$; useless when cold |
| $SO_2$ | 70% fossil fuel; residence 10-30 days; $H_2SO_4$ and $H_2SO_3$; FGD with $CaCO_3$; Pinatubo 25 Mt (1991); $H_2S_2O_6$ = dithionic |
| Smog | Industrial = $SO_2$ + soot, cold; photochemical = $NO_x$ + HC + sun; 1 $NO_2 \rightarrow$ 2 OH; 4 OH per HC; PAN = eye irritant; toluene fastest |
| Ozone | $O_2$ <240 nm → O; $O_3$ 220-330 nm; 20-30 km; 1 DU = $2.7\times10^{16}/cm^2$; $X + O_3 \rightarrow XO + O_2$, $XO + O \rightarrow X + O_2$; OH 40, NO 10,000, Cl 100,000 |